

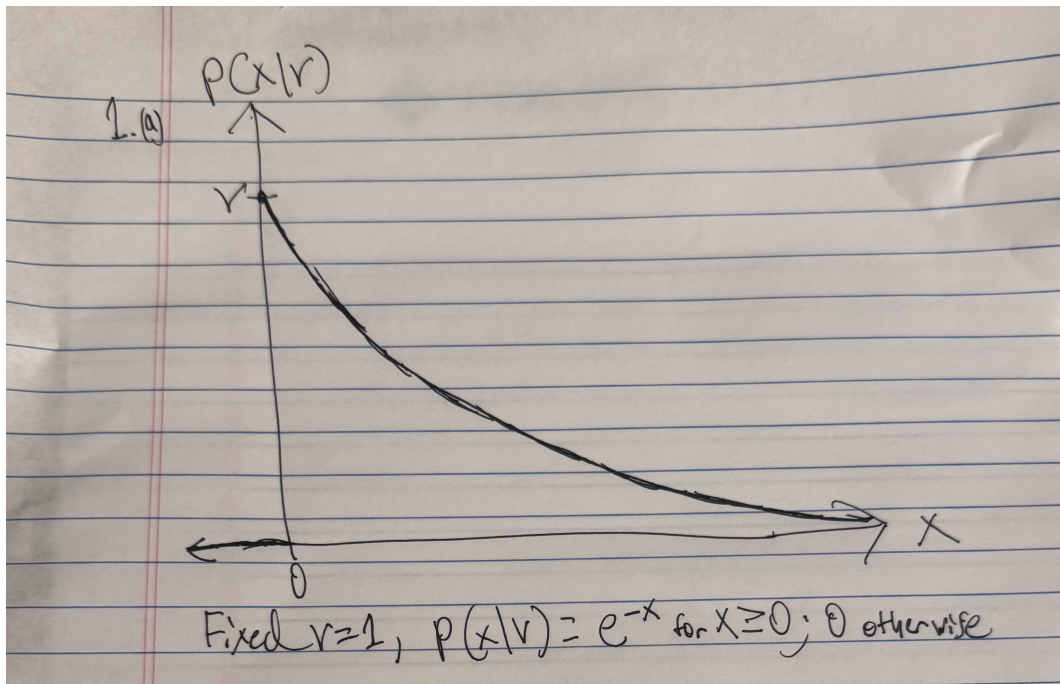
1.

Let  $X$  have an exponential distribution

$$p(x|v) = \begin{cases} ve^{-vx}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

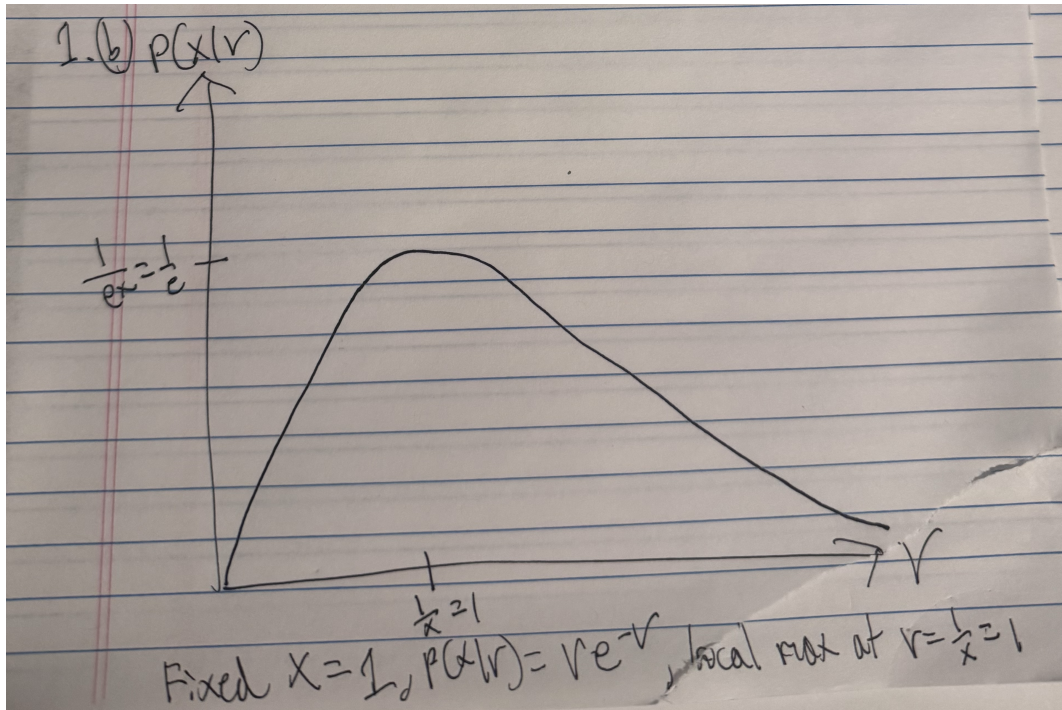
(a)

Sketch  $p(x|v)$  versus  $x$  for a fixed value of the parameter  $v$ .



(b)

Sketch  $p(x|v)$  versus  $v$ ,  $v > 0$  for a fixed value of  $x$ .



(c)

Suppose that  $n$  samples  $x_1, \dots, x_n$  are drawn independently according to  $p(x|v)$ . Show that the maximum likelihood estimate for  $v$  is given by

$$\hat{v} = \frac{1}{\frac{1}{n} \sum_{k=1}^n x_k}$$

We can derive the MLE for  $v$  directly. We first write the likelihood function. Suppose  $x_1, \dots, x_n$  are independent observations drawn from the exponential distribution, as defined by  $p(x|v)$ ; we just multiply for a likelihood function of

$$L(v) = \prod_{k=1}^n v e^{-v x_k}$$

We then take the log likelihood:

$$\begin{aligned} l(v) &= \ln\left(\prod_{k=1}^n v e^{-v x_k}\right) \\ &= \sum_{k=1}^n \ln(v e^{-v x_k}) \\ &= \sum_{k=1}^n (\ln(v) + \ln(e^{-v x_k})) \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=1}^n (\ln(v) - vx_k) \\
&= n \ln(v) - \sum_{k=1}^n vx_k
\end{aligned}$$

and differentiate:

$$\begin{aligned}
\frac{dl}{dv} &= \frac{d}{dv} (n \ln(v) - \sum_{k=1}^n vx_k) \\
&= \frac{n}{v} - \sum_{k=1}^n x_k
\end{aligned}$$

Verifying this is a maximum by taking the second derivative:

$$\begin{aligned}
\frac{d^2l}{dv^2} &= \frac{d}{dv} \left( \frac{n}{v} - \sum_{k=1}^n x_k \right) \\
&= -\frac{n}{v^2} - 0 = -\frac{n}{v^2}
\end{aligned}$$

which we see is indeed negative as both the numerator  $n > 0$  and the denominator is also always positive; multiplying by this then gives us a negative value, so this is indeed a maximum. Finally, set equal to zero to solve:

$$\begin{aligned}
0 &= \frac{n}{v} - \sum_{k=1}^n x_k \\
\sum_{k=1}^n x_k &= \frac{n}{v} \\
v &= \frac{n}{\sum_{k=1}^n x_k}
\end{aligned}$$

So we get

$$\hat{v} = \frac{n}{\sum_{k=1}^n x_k} = \frac{1}{\frac{1}{n} \sum_{k=1}^n x_k}$$

exactly as we sought to find.

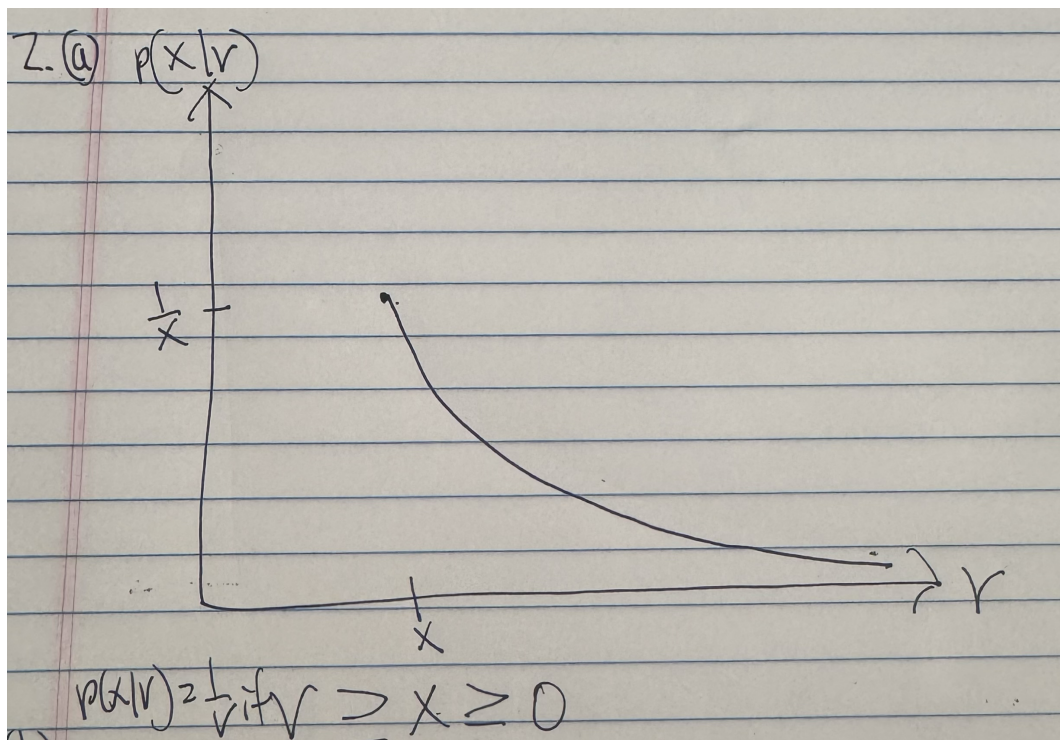
## 2.

Let  $X$  have a uniform distribution

$$p(x|v) = \begin{cases} \frac{1}{v}, & 0 \leq x \leq v \\ 0, & \text{otherwise.} \end{cases}$$

(a)

Sketch  $p(x|v)$  versus  $v$  for an arbitrary value of  $x$ .



(b)

Suppose that  $n$  samples  $x_1, \dots, x_n$  are drawn independently according to  $p(x|v)$ . Show that the maximum likelihood estimate for  $v$  is  $\max_k x_k$ .

By definition, we have that

$$L(v) = \prod_{k=1}^n p(x_k|v)$$
$$L(v) = \begin{cases} \frac{1}{v^n}, & v \geq \max_k(x_k) \\ 0, & \text{otherwise.} \end{cases}$$

We note that  $\frac{1}{v^n}$  continuously decreases as  $v$  increases; we therefore must choose the smallest possible value of  $v$ . We also have the restriction that  $v$  must be at least as large as the maximum  $x_k$  value, or  $\max_k x_k$ . We therefore have that

$$\hat{v} = \max_k x_k$$

exactly as we sought to show.

(c)

Find the method of moments estimator for  $v$ .

Let  $X_1, \dots, X_n \sim \text{Uniform}(0, v)$ . We therefore have the first population moment as

$$\begin{aligned}\alpha_1 = E[X] &= \int_0^v x \cdot \frac{1}{v} dx \\ &= \frac{1}{v} \cdot \frac{x^2}{2} \Big|_0^v \\ &= \frac{1}{v} \cdot \frac{v^2}{2} - 0 \\ E[X] &= \frac{v}{2}\end{aligned}$$

We would then also have the corresponding first sample moment as

$$\hat{\alpha}_1 = \frac{1}{n} \sum_{k=1}^n x_k = \bar{x}$$

So we finally take the method of moments estimator to be

$$\frac{\hat{v}}{2} = \bar{x}$$

$$\boxed{\hat{v} = 2\bar{x}}$$

**3.**

Let  $\mathbf{X}$  be a binary  $(0, 1)$  vector with multivariate Bernoulli distribution

$$p(\mathbf{x}|\mathbf{v}) = \prod_{i=1}^d v_i^{x_i} (1 - v_i)^{1-x_i}$$

where  $\mathbf{v} = (v_1, \dots, v_d)^T$  is an unknown parameter vector,  $v_i$  being the probability that  $x_i = 1$ . Show that the maximum likelihood estimate for  $\mathbf{v}$  is

$$\hat{\mathbf{v}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

We first evaluate the likelihood function, taking the product of all observations given  $n$  independent observations of  $\mathbf{x}$ :

$$L(\mathbf{v}) = \prod_{k=1}^n p(\mathbf{x}_k | \mathbf{v}) = \prod_{k=1}^n \prod_{i=1}^d v_i^{x_{ki}} (1 - v_i)^{1-x_{ki}}$$

We then take the log likelihood:

$$\begin{aligned} l(\mathbf{v}) &= \log\left(\prod_{k=1}^n \prod_{i=1}^d v_i^{x_{ki}} (1 - v_i)^{1-x_{ki}}\right) \\ &= \sum_{k=1}^n \sum_{i=1}^d \log(v_i^{x_{ki}} (1 - v_i)^{1-x_{ki}}) \\ l(\mathbf{v}) &= \sum_{k=1}^n \sum_{i=1}^d (x_{ki} \log(v_i) + (1 - x_{ki}) \log(1 - v_i)) \end{aligned}$$

Now taking the derivative of this:

$$\begin{aligned} \frac{\partial l}{\partial v_i} &= \frac{\partial}{\partial v_i} \left( \sum_{k=1}^n \sum_{i=1}^d (x_{ki} \log(v_i) + (1 - x_{ki}) \log(1 - v_i)) \right) \\ \frac{\partial l}{\partial v_i} &= \sum_{k=1}^n \sum_{i=1}^d \frac{\partial}{\partial v_i} (x_{ki} \log(v_i) + (1 - x_{ki}) \log(1 - v_i)) \end{aligned}$$

As we are only differentiating with respect to  $v_i$ , everything within the inner sum becomes 0 except for the  $v_i$  term itself, so this essentially simplifies to

$$\begin{aligned} \frac{\partial l}{\partial v_i} &= \sum_{k=1}^n \frac{\partial}{\partial v_i} (x_{ki} \log(v_i) + (1 - x_{ki}) \log(1 - v_i)) \\ \frac{\partial l}{\partial v_i} &= \sum_{k=1}^n \left( \frac{x_{ki}}{v_i} - \frac{1 - x_{ki}}{1 - v_i} \right) \end{aligned}$$

Now setting to 0 and solving:

$$\begin{aligned} 0 &= \sum_{k=1}^n \frac{x_{ki}}{v_i} - \sum_{k=1}^n \frac{1 - x_{ki}}{1 - v_i} \\ \sum_{k=1}^n \frac{x_{ki}}{v_i} &= \sum_{k=1}^n \frac{1 - x_{ki}}{1 - v_i} \end{aligned}$$

$$\begin{aligned}
\frac{1}{v_i} \sum_{k=1}^n x_{ki} &= \frac{1}{1-v_i} \sum_{k=1}^n (1-x_{ki}) \\
\frac{1}{v_i} \sum_{k=1}^n x_{ki} &= \frac{1}{1-v_i} (n - \sum_{k=1}^n x_{ki}) \\
(1-v_i) \sum_{k=1}^n x_{ki} &= v_i (n - \sum_{k=1}^n x_{ki}) \\
\sum_{k=1}^n x_{ki} - v_i \sum_{k=1}^n x_{ki} &= nv_i - v_i \sum_{k=1}^n x_{ki} \\
\sum_{k=1}^n x_{ki} &= nv_i \\
v_i &= \frac{1}{n} \sum_{k=1}^n x_{ki}
\end{aligned}$$

So combining this into the overall vector  $\mathbf{v}$ , we have that

$$\hat{\mathbf{v}} = \frac{1}{n} \sum_{k=1}^n \mathbf{x}_k$$

4.

Let  $X$  have a gamma distribution  $(\Gamma(x; \alpha, \beta))$

$$p(x|\alpha, \beta) = \begin{cases} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, & x > 0 \text{ and } \alpha, \beta > 0 \\ 0, & \text{otherwise.} \end{cases}$$

(a)

Suppose that  $n$  samples  $x_1, \dots, x_n$  are drawn independently according to  $p(x|\alpha, \beta)$ . Find the method of moments estimator for  $\alpha$  and  $\beta$ .

Because we have two unknown parameters we want to solve for, we take the first two moments. For the first population moment:

$$\alpha_1 = E[X] = \alpha\beta$$

And for the second population moment:

$$\begin{aligned}
\alpha_2 &= E[X^2] = \text{Var}(X) + E[X]^2 \\
&= \alpha\beta^2 + (\alpha\beta)^2 \\
&= \alpha\beta^2 + \alpha^2\beta^2 \\
\alpha_2 &= \alpha\beta^2(1 + \alpha)
\end{aligned}$$

Next, taking the sample moments:

$$\begin{aligned}
\hat{\alpha}_1 &= \frac{1}{n} \sum_{i=1}^n x_k \\
\hat{\alpha}_1 &= \bar{x}
\end{aligned}$$

And subsequently:

$$\hat{\alpha}_2 = \frac{1}{n} \sum_{i=1}^n x_k^2$$

We can now set up our equations and solve:

$$\alpha\beta = \bar{x}\alpha\beta^2(1 + \alpha) = \frac{1}{n} \sum_{i=1}^n x_k^2$$

Taking  $\beta = \frac{\bar{x}}{\alpha}$  and substituting:

$$\begin{aligned}
\alpha\left(\frac{\bar{x}}{\alpha}\right)^2(1 + \alpha) &= \frac{1}{n} \sum_{i=1}^n x_k^2 \\
\frac{\bar{x}^2(1 + \alpha)}{\alpha} &= \frac{1}{n} \sum_{i=1}^n x_k^2 \\
\bar{x}^2\left(1 + \frac{1}{\alpha}\right) &= \frac{1}{n} \sum_{i=1}^n x_k^2 \\
1 + \frac{1}{\alpha} &= \frac{\frac{1}{n} \sum_{i=1}^n x_k^2}{\bar{x}^2} \\
\frac{1}{\alpha} &= \frac{\frac{1}{n} \sum_{i=1}^n x_k^2 - \bar{x}^2}{\bar{x}^2}
\end{aligned}$$

We note that  $\frac{1}{n} \sum_{i=1}^n x_k^2 - \bar{x}^2$  is exactly the sample variance  $s^2$ , so therefore

$$\frac{1}{\alpha} = \frac{s^2}{\bar{x}^2}$$

$$\alpha = \frac{\bar{x}^2}{s^2}$$

And now solving for  $\beta$ :

$$\beta = \frac{\bar{x}}{\alpha} = \frac{\bar{x}}{\frac{\bar{x}^2}{s^2}} = \frac{s^2}{\bar{x}}$$

We therefore find that

$$\boxed{\hat{\alpha} = \frac{\bar{x}^2}{s^2}, \hat{\beta} = \frac{s^2}{\bar{x}}}$$

(b)

Show that the exponential distribution is  $\Gamma(x; 1, 1/v)$ .

We aim to essentially show that the exponential distribution pdf is

$$\begin{aligned} p(x|1, \frac{1}{v}) &= \frac{1}{\Gamma(1)^{\frac{1}{v}}_1} x^{1-1} e^{-x/\frac{1}{v}} \\ &= \frac{1}{0! \cdot \frac{1}{v}} \cdot x^0 \cdot e^{-vx} \\ &= v \cdot 1 \cdot e^{-vx} \\ &= v e^{-vx} \end{aligned}$$

which is exactly the standard exponential distribution with  $\lambda = v$ .

5.

6.

(a)

(b)

(c)